

ECE 341 Homework 2

The problem numbers listed below are from the Lathi textbook, 3rd edition.

Prob. Number	Course Outcome	Points
2.2-1 (b, d)	1.b	20
2.3-2	1.b	20
2.4-18 *	1.b	10
2.4-27 (b, f)	1.b	30
2.5-1	7.c	20
TOTAL:		100

*** Hint: You can use Table 2.1 of the textbook (Page 176) to find the convolution.**

1-b. Evaluate and classify signals to identify unstable systems.

7-c. Apply new knowledge to understand continuous and discrete systems.

2.2.1 (b) Given the system described by

$$\ddot{y}(t) + 2\dot{y}(t) + 5y(t) = \ddot{x}(t) - 5x(t),$$

with initial conditions

$$\begin{aligned} y_{\text{zir}}(0) &= 4 \\ \dot{y}_{\text{zir}}(0) &= -1, \end{aligned}$$

begin by setting $P(D)x(t)$ to zero to find the zero-input response, $y_{\text{zir}}(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t}$. To find λ_1 and λ_2 , solve the characteristic equation $\lambda^2 + 2\lambda + 5 = 0$, which returns two complex values,

$$\lambda_1 = -1 + j2, \lambda_2 = -1 - j2.$$

We can pick two way to solve for c_1 and c_2 : we can keep the equation in complex form and solve for the constants, or we can assume that the given system is real and therefore must have a real ZIR, which will allow us to solve for the constants c_1 and c_2 more easily. If assume that the ZIR is real, the c_1 and c_2 must be conjugates, expressed as

$$\begin{aligned} c_1 &= \frac{c}{2} e^{j\theta} \\ c_2 &= \frac{c}{2} e^{-j\theta}. \end{aligned}$$

Substituting these new constants into our expression for y_{zir} ,

$$y_{\text{zir}}(t) = \frac{c}{2} e^{j\theta} e^{(-1+j2)t} + \frac{c}{2} e^{-j\theta} e^{(-1-j2)t} = \frac{c}{2} [e^{-t} e^{j(2t+\theta)} + e^{-t} e^{j(-2t-\theta)}].$$

Simplifying using Euler's formula, $e^{jx} = \cos(x) + j \sin(x)$,

$$y_{\text{zir}}(t) = \frac{ce^{-t}}{2} [\cos(2t + \theta) + j \sin(2t + \theta) + \cos(-2t - \theta) + j \sin(-2t - \theta)].$$

Since $\cos(-x) = \cos(x)$ and $\sin(-x) = -\sin(x)$, this reduces to

$$\frac{ce^{-t}}{2} (2 \cos(2t + \theta)) = ce^{-t} \cos(2t + \theta).$$

Now, we can easily solve for c and θ in the real domain by setting $t = 0$ for $y_{\text{zir}}(t)$ and $\dot{y}_{\text{zir}}(t)$,

$$\begin{aligned} y_{\text{zir}}(t) &= ce^{-t} \cos(2t + \theta) \\ \dot{y}_{\text{zir}}(t) &= -ce^{-t} \cos(2t + \theta) - 2ce^{-t} \sin(2t + \theta). \end{aligned}$$

Setting these expressions equal to the given initial conditions with $t = 0$,

$$\begin{aligned} y_{\text{zir}}(0) &= c \cos(\theta) = 4 \\ \dot{y}_{\text{zir}}(0) &= c \cos(\theta) - 2c \sin(\theta) = -1. \end{aligned}$$

Solving for $c \cos(\theta)$ and $c \sin(\theta)$ using a system of equations,

$$\begin{bmatrix} 1 & 0 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} c \cos(\theta) \\ c \sin(\theta) \end{bmatrix} = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 4 \\ 1 & -2 & -1 \end{bmatrix} \Rightarrow \begin{aligned} c \cos(\theta) &= 4 \\ c \sin(\theta) &= 1.5 \end{aligned}$$

Considering both terms as the two coordinates of a complex number, we may solve for c and θ as $c = \sqrt{4^2 + 1.5^2} = 4.272$ and $\theta = \tan^{-1} \frac{1.5}{4} = 0.359$, we can substitute this back into our solution for $y_{\text{zir}}(t)$, which gives us the complete ZIR,

$$y_{\text{zir}}(t) = 4.272e^{-t} \cos(2t + 0.359).$$

Finally, we can convert our values for c and θ back to c_1 and c_2 if we want, as

$$c_1 = \frac{c}{2} e^{j\theta} = 2.136e^{j0.359}$$

$$c_2 = \frac{c}{2} e^{-j\theta} = 2.136e^{-j0.359}.$$

Then the equivalent, complex ZIR would be

$$y_{\text{zir}}(t) = 2.136e^{(-1+j0.359)t} + 2.136e^{(-1-j0.359)t}.$$

As mentioned, both equations are equivalent, and show an oscillatory, decaying response to the initial conditions, which is to be expected given complex-valued characteristic roots.

See Fig. 1 for a plot of the ZIR.

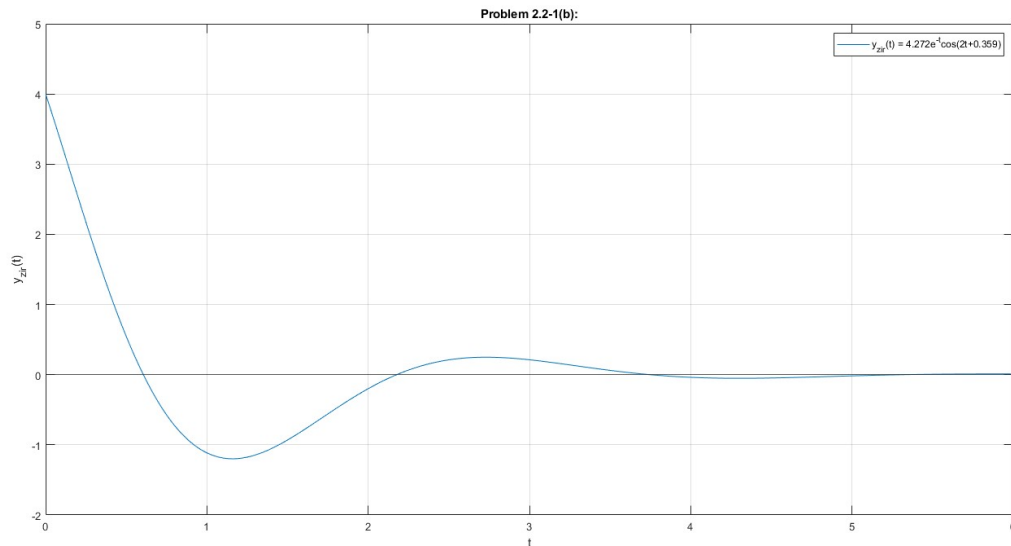


Figure 1: ZIR response